

Apparent Phantom Crossing as Template Bias: A Bounded-Clock Deformation of Λ CDM

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DESI DR2 reports a $2.8\text{--}4.2\sigma$ preference for dynamical dark energy in combined analyses, with standard w_0w_a fits favoring an apparent crossing of $w = -1$. We show that a non-phantom expansion history can produce apparent phantom crossings when analyzed with standard two-parameter templates. We construct a specific bounded-clock model, Λcos , and constrain it using Pantheon+ and DESI DR2 BAO. At data-allowed parameter values, the induced CPL distortion is structurally present but modest ($w_0 \approx -1.02$) and of opposite sign in w_a , establishing the mechanism without reproducing the DESI best-fit amplitude.

The Λcos model yields $H^2(z)/H_0^2 = \alpha(1+z)^3 - \beta(1+z) + \Omega_\Lambda$, where α and β are determined by a single parameter s_0 and a fixed reference value $\Omega_\Lambda = 0.685$. Under the fiducial-matter diagnostic split, the model satisfies $w_{\text{eff}}(z) > -1$ at all redshifts, recovers the fiducial flat Λ CDM limit as $s_0 \rightarrow 0$, and contains a negative $(1+z)^1$ term in H^2 absent from the four canonical FLRW density scalings. A joint MCMC fit to Pantheon+ (1701 SNe Ia) and DESI DR2 BAO (13 data points) yields $\Delta\chi^2 = +0.11$ relative to flat Λ CDM at the same parameter count, with $s_0 < 0.19$ (95% CL, flat prior). The constraint is robust across prior choices ($s_0 < 0.12\text{--}0.21$) and Ω_Λ variations (0.68–0.715). Adding compressed Planck distance priors shifts the preferred Ω_Λ in both Λ CDM and Λcos ; allowing Ω_Λ to vary gives statistically equivalent fits ($\Delta\chi^2 = +0.28$ at equal parameter count). A growth-level consistency check using DESI DR1 full-shape ShapeFit growth amplitudes gives $\Delta\chi_{\text{RSD}}^2 < 0.3$ relative to flat Λ CDM across the data-allowed parameter range.

I. INTRODUCTION

The DESI collaboration’s second data release [1] presents the most precise baryon acoustic oscillation measurements to date, spanning $0.3 < z < 2.3$ across seven tracer populations. Combined with Type Ia supernovae (Pantheon+ [2]) and CMB calibration (Planck [3]), the data favor a time-varying dark energy equation of state over the cosmological constant at $2.8\text{--}4.2\sigma$ within standard two-parameter parameterizations.

The inferred evolution follows the Chevallier-Polarski-Linder (CPL) parameterization $w(a) = w_0 + w_a(1-a)$ [4, 5], with best-fit values suggesting $w_0 \approx -0.75$ and $w_a \approx -0.86$. This implies a crossing of $w = -1$ near $z \approx 0.5$. In minimally coupled canonical single-field dark energy, crossing $w = -1$ is forbidden; realizing such behavior generally requires additional structure, such as noncanonical dynamics, multiple fields, interactions, or modified gravity [6, 7]. Cort  s and Liddle [8] flag the crossing location at the center of the observable window as a “substantial and unsettling coincidence.” The DESI extended analysis [9] reports consistent phantom-crossing behavior across five additional parameterizations (BA [10], JBP [11], exponential, logarithmic, and model-free binned).

The possibility that apparent phantom crossings can arise from parameterization choice has been recognized. Linder and Huterer [12] demonstrated that the CPL form introduces systematic bias when the true $w(z)$ lies outside its functional family. Shafieloo, Sahni, and Starobinsky [13] showed that model-dependent reconstruction can

produce spurious features, including phantom crossings, when applied to models outside the fitting basis. More recently, several groups have examined whether the DESI signal specifically could reflect such artifacts [8, 14].

The key question is not whether phantom crossings can be fitted to current data, but whether they are physically required, or instead arise as artifacts of the fitting template applied to a non-phantom expansion history. The present paper contributes to this discussion in two ways.

First, we construct a specific bounded-clock model (Λcos), a one-parameter deformation of fiducial flat Λ CDM whose diagnostic effective residual remains non-phantom under the fiducial-matter split, providing a falsifiable model rather than a generic cautionary argument. The model yields $H^2(z)/H_0^2 = \alpha(1+z)^3 - \beta(1+z) + \Omega_\Lambda$, for which $w_{\text{eff}}(z) > -1$ follows analytically at all redshifts (Sec. III), and which recovers the fiducial flat Λ CDM limit as its single new parameter $s_0 \rightarrow 0$.

Second, we test the template bias mechanism quantitatively: we show that CPL, BA, and JBP all produce apparent phantom crossings when fitted to Λcos distances, and we scan the full parameter range to determine the amplitude of the induced distortion at data-allowed values. At the 95% upper limit $s_0 < 0.19$ from a joint Pantheon+ and DESI DR2 BAO fit, the CPL recovery gives $w_0 \approx -1.02$ with positive w_a , establishing the structural mechanism while falling short of the DESI best-fit amplitude and sign.

The goal is not to claim that Λcos explains the full DESI preference, but to provide a specific bounded-clock model in which the size and sign of template-induced phantom behavior can be computed and confronted with current data.

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The paper is organized as follows. Sec. II constructs the ACOS model from a one-parameter bounded-clock construction and selects the clock exponent by matter-era scaling. Sec. III proves $w_{\text{eff}} > -1$ at all redshifts. Sec. IV demonstrates the template bias mechanism and scans its amplitude. Sec. V presents observational constraints including prior sensitivity, Ω_Λ robustness, and CMB distance priors. Sec. VI collects predictions and falsification criteria. Sec. VII discusses the results.

II. THE ACOS MODEL

A. Background construction

We construct a one-parameter deformation of the fiducial flat Λ CDM expansion history using a bounded auxiliary variable $S \in (0, 1]$, related to redshift by $1 + z = s_0/S$, where s_0 is the present-epoch value of S . A bounded function of a monotonic phase variable t enters the bounded-clock family if it satisfies three structural requirements: S is bounded above by unity, $S \rightarrow 0$ at early times ($t \rightarrow 0$), and no additional shape parameter is introduced beyond s_0 . The choice $S = \sin(t/2)$ satisfies all three on the monotonic branch $0 < t \leq \pi$.

The redshift relation $1 + z = s_0/S$ reduces to the standard FLRW form $a/a_0 = S/s_0$ with the identification $a \propto S$. The Λ CDM limit corresponds to $s_0 \rightarrow 0$ with S/s_0 held fixed, in which the trigonometric corrections vanish and S/s_0 reduces to a conventional scale factor.

Any bounded function satisfying these three requirements defines a member of the bounded-clock family. Within the power-law clock subfamily, the exponent is not free: the matter-era Friedmann scaling $H^2 \propto (1+z)^3$ at high redshift uniquely selects $n = -1/2$ (Sec. II B), and all integer alternatives are ruled out empirically at $\Delta\chi^2 > 60$ (Appendix A). Among bounded functions compatible with the selected exponent, $S = \sin(t/2)$ yields a finite polynomial $H^2(z)$ in $(1+z)$ with coefficients determined by s_0 alone (Sec. II C), permitting an analytic proof that $w_{\text{eff}} > -1$ at all redshifts (Sec. III) and direct comparison with standard FLRW density scalings. Alternative bounded parameterizations satisfying the same constraints would generically yield transcendental, rational, or infinite-series expressions for $H^2(z)$, rather than the finite two-term polynomial obtained here.

B. Clock selection

The phase variable t and proper time τ are related by a clock rate. From $S = \sin(t/2)$, the non-vacuum scaling of the expansion rate is governed by:

$$\frac{1}{S} \frac{dS}{d\tau} = \frac{\cos(t/2)}{2 \sin(t/2)} \cdot \frac{dt}{d\tau}$$

This quantity is not yet the physical Hubble parameter; it defines the redshift scaling assigned to the non-vacuum sector before the cosmological constant is added (Sec. II C).

For a power-law clock $dt/d\tau = S^n$, the non-vacuum scaling gives $(1/S)(dS/d\tau) = \frac{1}{2} \cos(t/2) S^{n-1}$. The power-law form is the minimal one-parameter family relating phase and proper time; more general clocks $dt/d\tau = f(S)$ would introduce functional freedom beyond a single exponent and are not considered here. Consistency with the observed matter-dominated expansion at high redshift requires $H^2 \propto \rho_m \propto S^{-3}$ [15], which uniquely selects the clock exponent within the power-law family. At high redshift ($S \rightarrow 0$), $\cos(t/2) \rightarrow 1$, and the requirement $(1/S)(dS/d\tau) \propto S^{-3/2}$ selects $n - 1 = -3/2$, giving:

$$\frac{dt}{d\tau} = S^{-1/2} = \sin^{-1/2}(t/2)$$

With this clock, the exact non-vacuum kernel is:

$$\frac{1}{S} \frac{dS}{d\tau} = \frac{\sqrt{1-S^2}}{2S^{3/2}}$$

The $\sqrt{1-S^2}$ factor equals unity at high z and generates the $(1+z)^1$ correction at low z (Sec. II D). For the tested integer alternatives, $n = +1, 0, -1$ give $(1/S)(dS/d\tau) \propto (1+z)^0, (1+z)^1, (1+z)^2$, respectively. Only $n = -1/2$ gives the matter-era scaling $(1+z)^3/2$. This was verified against Pantheon+ data, where all three integer-power alternatives give $\Delta\chi^2 > 60$ relative to Λ CDM (Appendix A).

C. The Hubble rate

Using $1 + z = s_0/S$, we write $S = s_0/(1+z)$. Squaring the non-vacuum kernel from Sec. II B and substituting:

$$\mathcal{K}(z; s_0) = \frac{(1+z)^3 - s_0^2(1+z)}{1 - s_0^2}$$

where the normalization $\mathcal{K}(0; s_0) = 1$ is enforced by the denominator. The cosmological constant, as a vacuum energy density independent of the expansion history, enters the Friedmann equation [15] as a separate additive component. The physical dimensionless Hubble rate is then:

$$E^2(z) \equiv \frac{H^2(z)}{H_0^2} = (1 - \Omega_\Lambda) \mathcal{K}(z; s_0) + \Omega_\Lambda$$

$\Omega_\Lambda = 0.685$ is taken as a fixed reference value; Sec. V D demonstrates stability of all results across the range 0.68–0.715.

Expanding:

$$\frac{H^2(z)}{H_0^2} = \frac{1 - \Omega_\Lambda}{1 - s_0^2} (1+z)^3 - \frac{(1 - \Omega_\Lambda) s_0^2}{1 - s_0^2} (1+z) + \Omega_\Lambda$$

Defining $\alpha \equiv (1 - \Omega_\Lambda)/(1 - s_0^2)$ and $\beta \equiv (1 - \Omega_\Lambda)s_0^2/(1 - s_0^2)$, the expression reads $H^2/H_0^2 = \alpha(1+z)^3 - \beta(1+z) + \Omega_\Lambda$, with $\alpha - \beta + \Omega_\Lambda = 1$ enforcing $E(0) = 1$. Radiation ($\Omega_r \approx 9 \times 10^{-5}$) is omitted; its contribution is below 0.3% at all redshifts probed by the datasets used here ($z < 2.4$).

In the baseline fits, Ω_Λ is fixed to the reference value above, so the cosmological degree of freedom replacing Ω_m is s_0 . The baseline Λ cos and flat Λ CDM comparisons therefore use the same number of fitted parameters: s_0 , $H_0 r_d$, and M_B for Λ cos; Ω_m , $H_0 r_d$, and M_B for flat Λ CDM. The fiducial flat Λ CDM expansion history is recovered as $s_0 \rightarrow 0$, where $\alpha \rightarrow 1 - \Omega_\Lambda$ and $\beta \rightarrow 0$.

D. The $(1+z)^1$ term

The negative $(1+z)^1$ term is absent from the four canonical FLRW components:

TABLE I. FLRW density components compared with the bounded $(1+z)^1$ correction term in Λ cos.

Component	Redshift scaling
Radiation	$(1+z)^4$
Matter	$(1+z)^3$
Curvature	$(1+z)^2$
Cosmological constant	$(1+z)^0$
Λcos correction	$(1+z)^1$

A constant- w fluid with $w = -2/3$ (domain walls) would produce the same redshift scaling. The Λ cos term is distinguished here by the tied coefficient $-\beta(s_0)$, rather than by the scaling alone. Its coefficient is $-\beta = -(1 - \Omega_\Lambda)s_0^2/(1 - s_0^2)$, strictly negative for $s_0 > 0$ and vanishing in the Λ CDM limit. Interpreted as an effective fluid, the negative coefficient would imply a violation of the weak energy condition. However, the $(1+z)^1$ term is not an independent component but a correction arising from the bounded parameterization of the matter sector; energy condition arguments apply to physical stress-energy, not to diagnostic residuals of an algebraic decomposition.

III. $w_{\text{eff}}(\mathbf{Z}) > -1$ AT ALL REDSHIFTS

To define a diagnostic effective dark energy equation of state, one must specify a matter subtraction. We adopt the split in which deviations from the fiducial matter scaling are assigned to the effective dark-energy sector, using $\Omega_m = 1 - \Omega_\Lambda = 0.315$ (the fiducial matter fraction, independent of s_0) rather than the dressed coefficient $\alpha = \Omega_m/(1 - s_0^2)$ that appears in the $(1+z)^3$ term.

The effective dark energy density is then:

$$\begin{aligned} X(z) &= E^2(z) - \Omega_m(1+z)^3 \\ &= \frac{\Omega_m s_0^2}{1 - s_0^2} [(1+z)^3 - (1+z)] + \Omega_\Lambda \end{aligned}$$

Treating this effective component as if it satisfies the standard continuity equation,

$$d\rho_{\text{DE}}/dz = 3(1 + w_{\text{eff}})\rho_{\text{DE}}/(1+z),$$

defines a diagnostic mapping from $H(z)$ to an effective equation-of-state parameter. This is not a commitment to $X(z)$ being a physical fluid; the $(1+z)^1$ term is a background-level correction rather than an independent component (Sec. IID). The diagnostic equation of state is:

$$w_{\text{eff}}(z) = -1 + \frac{(1+z)}{3X} \frac{dX}{dz}$$

Computing the derivative:

$$\frac{dX}{dz} = \frac{\Omega_m s_0^2}{1 - s_0^2} [3(1+z)^2 - 1]$$

The correction to $w = -1$ is therefore:

$$w_{\text{eff}}(z) + 1 = \frac{\Omega_m s_0^2 (1+z) [3(1+z)^2 - 1]}{3(1 - s_0^2) \left[\frac{\Omega_m s_0^2}{1 - s_0^2} ((1+z)^3 - (1+z)) + \Omega_\Lambda \right]}$$

For $z \geq 0$ (corresponding to $S \leq s_0$) and $s_0 \in (0, 1)$:

- **Numerator:** $3(1+z)^2 - 1 \geq 2$ (equality at $z = 0$). Strictly positive.
- **Denominator:** $(1+z)^3 - (1+z) = (1+z)(z)(2+z) \geq 0$ for $z \geq 0$, and the additive $\Omega_\Lambda > 0$ ensures $X(z)$ is strictly positive at every z .

Therefore:

$$w_{\text{eff}}(z) > -1$$

for the fiducial-matter diagnostic split, for all $z \geq 0$, $s_0 \in (0, 1)$.

The constraint $0 < s_0 < 1$ follows from the definition $s_0 = \sin(t_{\text{now}}/2)$ with $0 < t_{\text{now}} \leq \pi$ (Sec. IIA). At $s_0 = 0$, $w_{\text{eff}} = -1$ everywhere (the fiducial flat Λ CDM limit). For $s_0 > 0$, the correction is positive and begins at order s_0^2 .

The non-phantom classification is diagnostic and split-dependent by construction. The fiducial split (subtracting $\Omega_m = 1 - \Omega_\Lambda = 0.315$, independent of s_0) assigns the s_0 -dependent correction to the effective dark-energy sector, yielding $w_{\text{eff}} > -1$ at all redshifts. The dressed split (subtracting $\alpha(1+z)^3$) assigns the correction to the

matter sector, yielding $w_{\text{eff}} < -1$ at high z . Both are valid decompositions of the same expansion history; neither changes the physical distances or the Hubble rate. This paper adopts the fiducial split because it holds the matter fraction fixed as s_0 varies, making the diagnostic equation of state a function of the single new parameter alone. The convention-dependence is itself part of the template bias finding (Sec. VII A): the same $H(z)$ is diagnosed as phantom or non-phantom depending on the subtraction, which is the ambiguity that two-parameter templates exploit.

IV. TEMPLATE BIAS DEMONSTRATION

A. Method

To test whether standard $w(z)$ parameterizations can produce apparent phantom crossings from Λcos distances, we generate noise-free BAO observables (D_M/r_d , D_H/r_d , D_V/r_d) from the Λcos model at the seven DESI DR2 effective redshifts ($z = 0.295, 0.510, 0.706, 0.934, 1.321, 1.484, 2.330$), using $H_0 = 67.4$ km/s/Mpc and $r_d = 147.1$ Mpc. We then fit four $w(z)$ parameterizations to these mock observables using the published DESI covariance structure, with $\Omega_m = 0.315$ held fixed to match the input Λcos model, isolating the effect of the $w(z)$ parameterization. For each effective redshift we use the same observable set and covariance block structure as the DESI DR2 compressed BAO likelihood; when multiple observables are reported at the same redshift, their published correlation coefficients are included.

The four parameterizations are CPL [4, 5], Barboza-Alcaniz (BA) [10], Jassal-Bagla-Padmanabhan (JBP) [11], and a three-parameter polynomial in (1-a). Each is a standard fitting form used in the DESI extended analysis [9].

The mock exercise uses $s_0 = 0.389$ to maximize visibility of the effect. The template bias mechanism operates at any $s_0 > 0$ (see Sec. IV C). The fits minimize $\chi^2 = \Delta^T C^{-1} \Delta$, where Δ is the difference between model and mock BAO observables and C is the DESI DR2 covariance matrix constructed from the published uncertainties and correlations. The χ^2 values reported below reflect fits to noise-free mock data with DESI covariance weighting.

B. Results

Three of four parameterizations produce phantom crossings from input that satisfies $w > -1$ everywhere (Sec. III). The polynomial, with one additional free parameter, fits the Λcos distances to $\chi^2 \approx 0$ on 13 BAO points and does not produce a phantom crossing: the apparent crossing in the two-parameter forms is purely a basis-restriction artifact, not a feature of the underlying expansion history. The effect arises from projecting the curvature of the Λcos distance-redshift relation onto

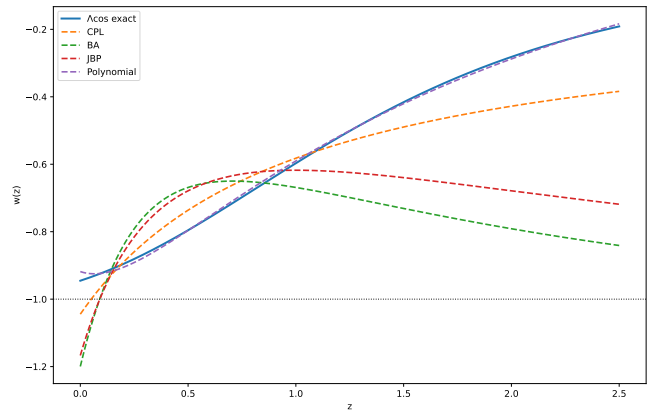


FIG. 1. Exact Λcos $w_{\text{eff}}(z)$ (solid, always > -1) overlaid with recovered $w(z)$ from all four parameterizations. CPL, BA, and JBP dip below $w = -1$ at low z . The three-parameter polynomial tracks the true curve without crossing.

a restricted two-parameter basis, absorbing the residual into phantom-crossing parameter values.

C. Threshold scan

To determine the amplitude of the template bias at data-allowed values of s_0 , we repeat the CPL fit across $s_0 \in [0.01, 0.40]$ in steps of 0.01. The CPL recovery produces an apparent phantom-crossing trajectory at every s_0 tested, including the smallest value $s_0 = 0.01$: the fitted $w_0 < -1$ at low redshift, with the positive w_a driving w back above -1 at higher redshift. Representative results:

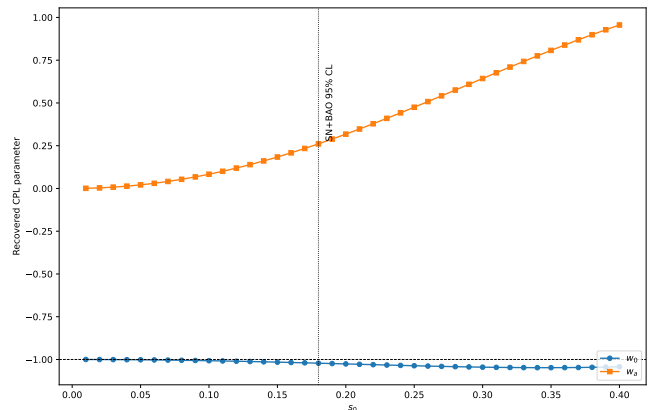


FIG. 2. Recovered CPL parameters w_0 (circles) and w_a (squares) as a function of s_0 . The apparent crossing trajectory persists at every tested $s_0 > 0$. The horizontal dashed line marks $w = -1$.

Two features are evident. First, the crossing persists at every tested $s_0 > 0$, confirming that the template bias is a structural property of the CPL basis applied to non-

TABLE II. Best-fit $w(z)$ parameters from CPL, BA, JBP, and a three-parameter polynomial fitted to noise-free Acos mock BAO at $s_0 = 0.389$.

Parameterization	Best-fit parameters	χ^2	Crosses $w = -1$?
CPL: $w_0 + w_a(1-a)$	$w_0 = -1.045, w_a = 0.925$	1.31	Yes
BA: $w_0 + w_a a(1-a)/(a^2-2a+2)$	$w_0 = -1.200, w_a = 2.654$	11.48	Yes
JBP: $w_0 + w_a a(1-a)$	$w_0 = -1.167, w_a = 2.197$	7.21	Yes
Polynomial: $w_0 + w_1(1-a) + w_2(1-a)^2$	$w_0 = -0.918, w_1 = -0.213, w_2 = 1.739$	0.004	No

TABLE III. Recovered CPL parameters as a function of s_0 , illustrating that the apparent crossing trajectory persists at every tested $s_0 > 0$.

s_0	w_0	w_a	w_0 deviation from -1
0.01	-1.00007	+0.001	7×10^{-5}
0.06	-1.003	+0.030	0.003
0.18	-1.022	+0.261	0.022
0.39	-1.044	+0.928	0.044
DESI observed	-0.75	-0.86	0.25

phantom models of this type, not an artifact of a particular parameter value. Second, at the SN+BAO 95% upper limit $s_0 < 0.19$ (Sec. V), the induced distortion is modest: $w_0 \approx -1.02$ with $w_a \approx +0.29$.

D. Comparison with the DESI best fit

The DESI CPL best fit reports $w_0 \approx -0.75$ and $w_a \approx -0.86$. The Acos-induced distortion at data-allowed s_0 differs in both amplitude ($|w_0 + 1| = 0.02$ vs 0.25, a factor of ~ 10) and sign ($w_a = +0.29$ vs -0.86).

The template bias mechanism is therefore established as a structural effect, but the Acos model at its current constraint does not reproduce the DESI best-fit parameters quantitatively. Bridging the amplitude gap would require either a larger bounded deformation ($s_0 \gtrsim 0.4$, ruled out by current data) or a different functional form for the effective dark-energy sector.

This distinction is important. The paper demonstrates that a non-phantom expansion history (Acos) produces apparent phantom crossings under standard two-parameter templates. The claim is not that Acos at $s_0 < 0.19$ explains the specific amplitude of the DESI signal.

V. OBSERVATIONAL CONSTRAINTS

A. Data

We fit jointly to the Pantheon+ supernova compilation (1701 SNe Ia with full statistical and systematic covariance [2]) and DESI DR2 BAO measurements (1 isotropic D_V/r_d at $z = 0.295$ and 6 anisotropic $D_M/r_d, D_H/r_d$ pairs at $z = 0.51-2.33$, with published cross-correlations [1]). The total dataset comprises 1714 observables (1701 supernova magnitudes and 13 BAO measurements).

The supernova likelihood is $\chi_{\text{SN}}^2 = \Delta^T C^{-1} \Delta$, where $\Delta_i = m_{b,i} - M_B - \mu(z_i)$ and C is the 1701×1701 covariance matrix. The BAO likelihood is constructed from the published uncertainties and inter-observable correlations at each redshift bin, with bins treated as independent. The total likelihood is $\chi^2 = \chi_{\text{SN}}^2 + \chi_{\text{BAO}}^2$. The growth comparison in Sec. VIC is not included in the primary SN+BAO likelihood; it uses DESI DR1 ShapeFit compressed growth amplitudes [16] at six tracer effective redshifts. Combining DESI DR2 BAO with DESI DR1 full-shape information follows the approach of [17, 18].

B. Primary fit: SN+BAO

MCMC sampling uses an affine-invariant ensemble sampler (32 walkers, 5000 steps, 1000 burn-in). Convergence is confirmed via integrated autocorrelation time ($\tau_{\text{max}} \approx 46$ for Acos and $\tau_{\text{max}} \approx 36$ for Λ CDM, with the post-burn chain length of $4000 \gtrsim 85\tau$ for all parameters).

Flat Λ CDM free parameters: $\Omega_m, H_0 r_d, M_B$. Acos free parameters: $s_0, H_0 r_d, M_B$ (with $\Omega_\Lambda = 0.685$ fixed). Both models have three free parameters. The primary results are reported at the fixed reference value $\Omega_\Lambda = 0.685$; Sec. VD demonstrates stability of the s_0 constraint across the range 0.68–0.715, and Sec. VE confirms the model remains competitive when Ω_Λ is freed entirely ($\Delta\chi^2 = +0.28$ at equal parameter count).

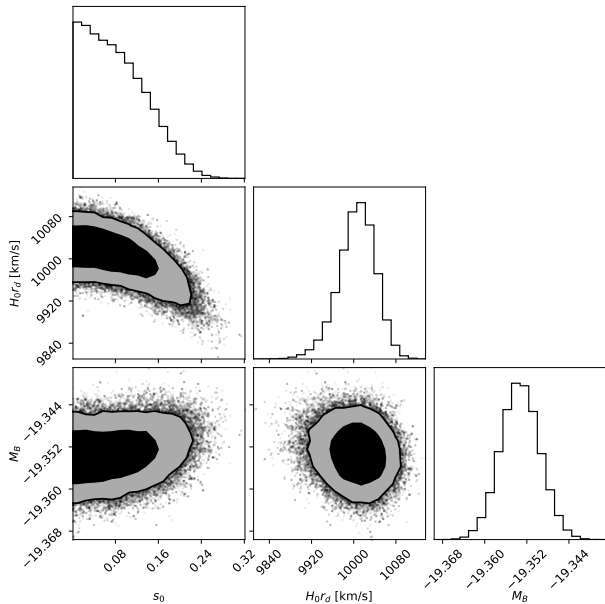
Uncertainties are posterior median and 68% credible intervals. The Acos best-fit sits at $s_0 = 0.001$ (the numerical lower bound of the prior; the analytic Λ CDM limit is $s_0 = 0$). The posterior median of 0.076 reflects prior volume at $s_0 > 0$. The 95th percentile gives:

$$s_0 < 0.19 \quad (95\% \text{ CL, flat prior}).$$

The $\Delta\chi^2 = +0.11$ at equal parameter count corresponds to $\Delta\text{AIC} = \Delta\text{BIC} = +0.11$: no preference for either model.

TABLE IV. Joint Pantheon+ + DESI DR2 BAO posterior summaries for flat Λ CDM and Λ cos.

	Flat Λ CDM	Λ cos
Primary parameter	$\Omega_m = 0.312$ [0.304, 0.321]	$s_0 = 0.076$ [0.023, 0.143]
$H_0 r_d$ (km/s)	10043 [9977, 10111]	10008 [9972, 10040]
M_B	-19.355 [-19.360, -19.350]	-19.353 [-19.357, -19.350]
$\chi^2_{\min}$	1772.5	1772.6
$\chi^2_{\text{SN}} / \chi^2_{\text{BAO}}$	1759.9 / 12.6	1759.0 / 13.5

FIG. 3. Corner plot of the Λ cos posterior in $(s_0, H_0 r_d, M_B)$ space from the SN+BAO fit.

C. Prior sensitivity

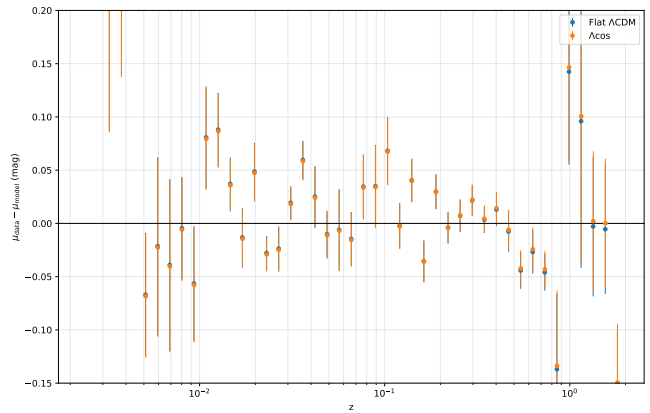
Since the Λ cos posterior is concentrated near the lower prior boundary, the 95% upper limit is prior-sensitive. We test two alternative priors by reweighting the baseline chain:

The $\chi^2_{\min}$ is identical across all three priors (the likelihood is unchanged; only the posterior weighting differs). The 95% upper limit ranges from 0.12 to 0.21, a factor of ~ 1.8 . The constraint is prior-sensitive in detail but data-driven in character: all three priors yield $s_0 \ll 1$.

D. Ω_Λ sensitivity

The primary fit uses $\Omega_\Lambda = 0.685$ as a fixed reference value. To test robustness, we repeat the SN+BAO fit at four alternative values spanning the range between the SN+BAO-preferred and CMB-preferred regions:

The s_0 constraint varies smoothly with Ω_Λ , with the 95% upper limit ranging from 0.16 to 0.35 across the tested interval. The best $\Delta\chi^2$ occurs near $\Omega_\Lambda = 0.69$. At

FIG. 4. Pantheon+ binned residuals ($\mu_{\text{data}} - \mu_{\text{model}}$) for Λ CDM and Λ cos at joint best-fit parameters. The two models are visually indistinguishable in the binned residuals.

the CMB-preferred value $\Omega_\Lambda = 0.715$, the best-fit s_0 shifts to 0.288 (nonzero) and the model is mildly disfavored ($\Delta\chi^2 = +2.34$). The results are stable across the tested range; no fine-tuning of Ω_Λ is required.

E. CMB distance priors

The DESI DR2 phantom-crossing result was obtained jointly with Planck CMB data. To test whether Λ cos is compatible with CMB constraints, we add compressed Planck 2018 distance priors (shift parameter $R = 1.7502 \pm 0.0046$, acoustic scale $\ell_A = 301.47 \pm 0.09$, treated as independent Gaussians) to the SN+BAO likelihood. These quantities constrain the integral $\int dz/E(z)$ to the last-scattering surface at $z^* = 1090$. Radiation ($\Omega_r = 9.15 \times 10^{-5}$) is included for this integral. Non-flat Λ CDM includes a curvature term $\Omega_k = 1 - \Omega_m - \Omega_\Lambda$. Since the Planck distance priors are derived within a restricted background model space, their use here should be understood as a consistency diagnostic rather than a full CMB likelihood evaluation.

The four-parameter Λ cos fit with Ω_Λ free and the four-parameter non-flat Λ CDM fit provide the primary comparison:

The two models provide statistically equivalent fits at $\Delta\chi^2 = +0.28$ with the same parameter count.

For comparison, fixing $\Omega_\Lambda = 0.685$ introduces a penalty of $\Delta\chi^2 \approx +66$ relative to flat Λ CDM (3 parameters each).

TABLE V. Prior sensitivity of the Λ cos s_0 constraint across three prior choices.

Prior on s_0	s_0 median	s_0 95% UL	$\chi^2_{\min}$
Flat $s_0 \in [0.001, 0.99]$ (baseline)	0.076	0.185	1772.6
Flat in s_0^2 (more weight at larger s_0)	0.119	0.211	1772.6
Flat in $\log_{10}(s_0)$ (scale-invariant)	0.012	0.119	1772.6

TABLE VI. Λ cos sensitivity to the Ω_Λ reference value across the SN+BAO and CMB-preferred range.

Ω_Λ	s_0 median	s_0 95% UL	$\Delta\chi^2$ vs Λ CDM
0.680	0.063	0.165	+0.83
0.685 (baseline)	0.076	0.185	+0.11
0.690	0.089	0.205	+0.07
0.700	0.147	0.260	+0.96
0.715 (CMB-preferred)	0.278	0.349	+2.34

TABLE VII. Comparison of Λ cos with Ω_Λ free against non-flat Λ CDM after adding compressed Planck distance priors.

	Λ cos (Ω_Λ free)	Non-flat Λ CDM
Free params $s_0, \Omega_\Lambda, H_0 r_d, M_B$	$\Omega_m, \Omega_\Lambda, H_0 r_d, M_B$	
Best-fit Ω_Λ	0.714 [0.711, 0.718]	0.720 [0.708, 0.732]
$\chi^2_{\min}$	1814.8	1814.5
$\Delta\chi^2$	+0.28	0 (reference)

This tension is driven by the BAO sector ($\chi^2_{\text{BAO}} = 120$ vs 30), not by the CMB priors themselves ($\chi^2_{\text{CMB}} = 1.3$ for Λ cos vs 14.7 for Λ CDM). The effect is not specific to Λ cos: adding CMB priors shifts the preferred Ω_Λ from 0.685–0.688 (SN+BAO) to ≈ 0.714 (Λ cos with Ω_Λ free) and ≈ 0.720 (non-flat Λ CDM). Flat Λ CDM absorbs the shift through Ω_m (which moves from 0.312 to 0.286). Λ cos at fixed Ω_Λ cannot absorb it. The shift from $\Omega_\Lambda \approx 0.685$ to $\Omega_\Lambda \approx 0.714$ should therefore be interpreted as a background-distance consistency issue in the compressed-prior setup, rather than as a failure mode unique to Λ cos.

The compressed-prior test used here is an approximate background-level consistency check, not a substitute for the full correlated Planck likelihood.

F. Parameter accounting

The matter fraction in Λ cos is set through $\Omega_m = 1 - \Omega_\Lambda$ at the chosen reference value. The $(1+z)^3$ coefficient $\alpha = \Omega_m/(1-s_0^2)$ is a dressed version of this input. As a consistency check: at any s_0 , $\alpha(1 - s_0^2)$ returns $\Omega_m = 0.315$. The dressing is algebraically self-consistent and does not introduce additional freedom.

G. Model comparison

To contextualize the Λ cos result, we compare against wCDM (constant w as a free parameter, 4 free parameters: $\Omega_m, w, H_0 r_d, M_B$) fitted to the same SN+BAO

dataset:

TABLE VIII. Model comparison against Pantheon+ + DESI DR2 BAO: flat Λ CDM, Λ cos at two Ω_Λ values, and wCDM.

Model	Free params	$\chi^2_{\min}$	$\Delta\chi^2$	ΔAIC	ΔBIC
Flat Λ CDM	3	1772.5	0	0	0
Λ cos ($\Omega_\Lambda = 0.685$)	3	1772.6	+0.11	+0.11	+0.11
Λ cos ($\Omega_\Lambda = 0.715$)	3	1774.8	+2.34	+2.34	+2.34
wCDM	4	1759.4	-13.1	-11.1	-5.7

The wCDM fit yields $w = -0.855$ [-0.89, -0.82] (68% CI), preferring a quintessence-like ($w > -1$) departure from Λ CDM at $\Delta\text{AIC} = -11.1$. This is the same qualitative signal underlying the DESI phantom-crossing result: the SN+BAO data favor some departure from $w = -1$. The distinction is that wCDM captures this as a constant shift toward $w > -1$, while CPL projects it into a crossing through $w = -1$. The Λ cos model, for which $w_{\text{eff}} > -1$ at all redshifts (Sec. III), is consistent with the wCDM direction but does not achieve the same χ^2 improvement because the data-allowed s_0 is too small to produce a detectable departure.

The wCDM preference for $w \approx -0.855$ identifies a real departure from Λ CDM that Λ cos at data-allowed s_0 values cannot capture. Bridging this gap requires either a larger bounded deformation than current data permit or a different functional form for the non-vacuum sector; the template bias mechanism demonstrated in Sec. IV is necessary context for interpreting which direction the data favor.

For the baseline flat prior in s_0 , a Savage-Dickey density ratio at the s_0 prior boundary (with boundary-reflected KDE) gives $B_{01} \approx 7.1$ (stable to ± 0.03 across bandwidths), corresponding to moderate evidence for Λ CDM over Λ cos on the Jeffreys scale. This Bayes factor should be interpreted conditional on the baseline prior choice. The result confirms the $\Delta\chi^2$ finding: current data do not require the model's correction term.

VI. PREDICTIONS AND FALSIFICATION CRITERIA

A. The $(1+z)^1$ signature

The joint SN+BAO fit constrains $|\beta| < 0.012$ at 95% CL (from $s_0 < 0.19$ with $\Omega_m = 0.315$). The predicted fractional contribution to $H^2(z)$ at $z = 1$ is:

$$\frac{|\beta|(1+z)}{E^2(z)} < 0.8\% \quad (95\% \text{ CL at } z = 1).$$

This is below the precision of current BAO measurements (DESI DR2: $\sim 2\text{--}3\%$ per bin in D_H/r_d [1]) but potentially within reach of the next generation. Euclid DR1 spectroscopic BAO ($z = 0.9\text{--}1.8$, four redshift bins) is forecast at $1\text{--}2\%$ per bin [19], marginal for detection. The discriminating regime is next-generation surveys projected to reach sub-percent per-bin precision [19, 20], where the $(1+z)^1$ signature approaches per-bin detectability for s_0 in the upper portion of the data-allowed range, with correlated bins improving aggregate sensitivity.

A statistically significant detection of a negative $(1+z)^1$ component in $H^2(z)$ would constitute evidence for a non-standard contribution to the expansion history. While a domain-wall fluid ($w = -2/3$) produces the same scaling, the Λcos term is distinguished by the tied coefficient $-\beta(s_0)$ rather than by an independent energy density. Previous template-bias analyses identify the mechanism generically; the present model commits to a specific functional form whose coefficient is determined by a single parameter already constrained by current data.

B. Falsification criteria

Table IX summarizes the three load-bearing Λcos predictions and their falsification conditions.

C. Growth predictions: consistency check

The Λcos construction modifies the background expansion rate through the bounded-clock non-vacuum kernel (Sec. IID), rather than by adding an independent clustering component. In the standard subhorizon growth equation, the modification therefore enters through $H(z)$ and the usual matter Poisson source term. No additional effective sound speed or anisotropic-stress parameter is introduced.

We solve

$$\delta'' + \left(2 + \frac{H'}{H}\right) \delta' - \frac{3}{2} \Omega_m(z) \delta = 0$$

in $x = \ln a$, where primes denote derivatives with respect to x and

$$\Omega_m(z) = \frac{\Omega_m(1+z)^3}{E^2(z)}$$

uses the fiducial clustering matter density $\Omega_m = 1 - \Omega_\Lambda$. We impose matter-dominated initial conditions at $z_{\text{init}} = 1000$ and use $\sigma_{8,0} = 0.81$ as the $z = 0$ normalization. As a sanity check, the ΛCDM solution agrees with the standard growth-index approximation $f(z) \simeq \Omega_m(z)^{0.55}$ to

within 0.50% over the redshift range used here. This is a consistency check at the fixed SN+BAO best-fit parameters, not a joint MCMC; for the rigorous joint treatment of DESI DR1 full-shape and DR2 BAO likelihoods including cross-covariance, see [17, 18].

The comparison data are the six DESI DR1 ShapeFit+BAO compressed $f\sigma_{88}$ amplitudes from [16], Appendix A (Eqs. A.13–A.24), at tracer effective redshifts $z_{\text{eff}} = 0.30$ (BGS), 0.51 (LRG1), 0.71 (LRG2), 0.92 (LRG3), 1.32 (ELG2), and 1.49 (QSO). DESI DR2 provides the BAO measurements used in Sec. V but does not include a DR2 full-shape data release; DR1 is the most recent DESI growth dataset. We compare directly to the DESI-reported ShapeFit growth amplitude $f\sigma_{88}$. Where the quantity is discussed in relation to the usual $f\sigma_8$, the interpretation follows the DESI fiducial sound-horizon and shape-compression conventions. We use the marginalized diagonal uncertainties as a consistency diagnostic rather than a precision model-selection likelihood.

At the SN+BAO best fit, $\chi^2_{\text{RSD}} = 4.64$ for flat ΛCDM , 4.68 for Λcos at the posterior median ($s_0 = 0.076$), and 4.90 at the 95% upper limit ($s_0 = 0.185$), giving $\Delta\chi^2_{\text{RSD}} = +0.04$ and $+0.26$ respectively over six data points. The $f\sigma_{88}$ deviation is below 1% per bin at all DR1 effective redshifts, well within the $9\text{--}23\%$ per-bin precision (Fig. 5). The $\Omega_m(z)$ trajectory (Fig. 6) shows a more distinctive separation at high redshift: a 2.9% split between Λcos and ΛCDM at $z = 2.3$, beyond the reach of current growth data ($z_{\text{max}} = 1.49$).

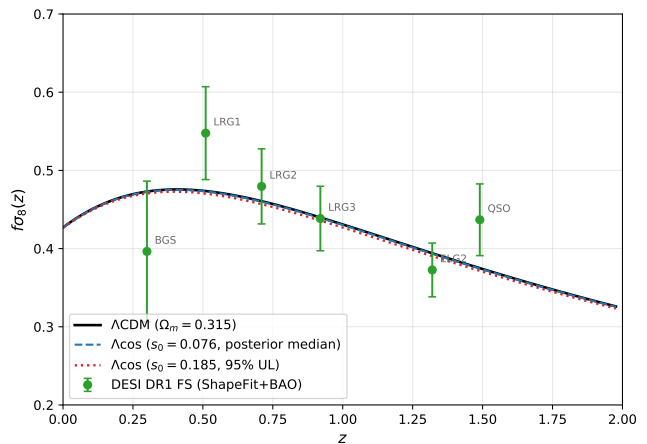


FIG. 5. ΛCDM (solid) and Λcos $f\sigma_{88}(z)$ trajectories at the posterior median ($s_0 = 0.076$, dashed) and 95% upper limit ($s_0 = 0.185$, dotted), overlaid with DESI DR1 ShapeFit+BAO growth amplitudes from [16]. The two models are visually indistinguishable at current per-bin precision.

Λcos passes the growth consistency check at current precision. Distinguishing the two models in growth observables requires next-generation full-shape analyses at sub-percent per-bin precision (Euclid DR2 spectroscopic, DESI full-survey) or higher-redshift growth mea-

TABLE IX. Falsification criteria for the three load-bearing Acos predictions.

Observable	Acos prediction	Falsified if
$(1+z)^1$ coefficient in H^2	Non-positive, magnitude tied to s_0	Positive coefficient detected, or negative coefficient inconsistent with fitted s_0
Diagnostic $w_{\text{eff}}(z)$	> -1 (fiducial split, any $s_0 > 0$)	Same split gives $w_{\text{eff}} < -1$ at $\geq 2\sigma$
Λ CDM limit	$s_0 = 0$ recovers fiducial flat Λ CDM	(none required)

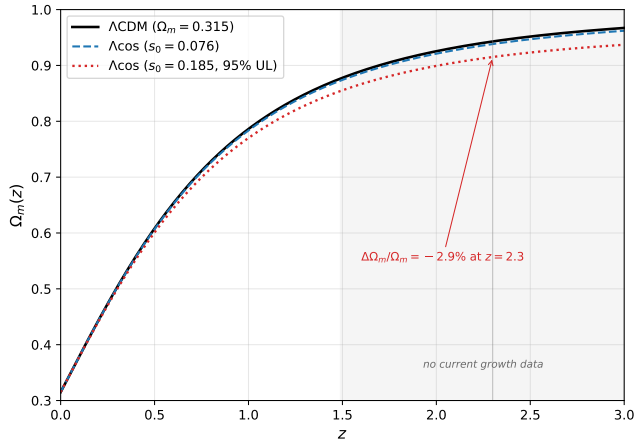


FIG. 6. $\Omega_m(z)$ diagnostic for Λ CDM and Acos at the 95% upper limit ($s_0 = 0.185$). The 2.9% split at $z = 2.3$ is beyond the reach of current growth data but accessible to next-generation surveys.

surements extending past $z = 1.49$. This parallels the conclusion from the $(1+z)^1$ BAO signature in Sec. VI A: the model is consistent with current data and falsifiable by the next generation.

VII. DISCUSSION

A. Template bias

The central result of this paper is that a non-phantom expansion history can produce apparent phantom crossings when fitted with standard two-parameter $w(z)$ templates. We have demonstrated this explicitly for CPL, BA, and JBP parameterizations applied to Acos distances: all three recover $w_0 < -1$ from Acos distances for which $w_{\text{eff}} > -1$ at every redshift under the fiducial-matter diagnostic split (Sec. III). A three-parameter polynomial does not produce the crossing, confirming that the effect arises from basis restriction rather than from the underlying expansion history.

This finding has precedent. Linder and Huterer [12] identified CPL template bias, Shafieloo, Sahni, and Starobinsky [13] demonstrated spurious phantom crossings from model-dependent reconstruction, and recent work [8, 14] has raised similar concerns about the DESI signal specifically. The present contribution is to provide a one-parameter non-phantom model (rather than a

generic cautionary argument), to test multiple parameterizations, and to scan the template bias amplitude across the full parameter range allowed by current data.

The DESI extended analysis [9] itself tests five parameterizations beyond CPL and finds consistent phantom-crossing behavior across all of them. This might seem to rule out template bias as an explanation. However, the DESI analysis fits each parameterization to real data, which may contain a genuine departure from Λ CDM (as the w CDM comparison in Sec. V G suggests). The mock exercise in Sec. IV asks a different question: can a model that provably satisfies $w > -1$ produce phantom crossings across the same family of templates? The answer is yes, confirming that template agreement across parameterizations is necessary but not sufficient evidence for a true crossing.

At data-allowed values ($s_0 < 0.19$), the induced CPL distortion is $w_0 \approx -1.02$ with positive w_a , structurally present but modest in amplitude and opposite in sign to the DESI best fit ($w_0 \approx -0.75$, $w_a \approx -0.86$). The paper establishes the mechanism without claiming to reproduce the DESI signal quantitatively.

B. Status of Acos

The Acos model is fixed by two structural constraints and one reference normalization: a bounded auxiliary variable $S = \sin(t/2)$ with redshift relation $1+z = s_0/S$ (Sec. II A), the power-law clock exponent $n = -1/2$ selected by matter-era Friedmann scaling (Sec. II B), and a reference value $\Omega_\Lambda = 0.685$ (Sec. II C). The resulting $H^2(z)$ is derived algebraically from these inputs. The fiducial flat Λ CDM expansion history is recovered as $s_0 \rightarrow 0$.

The Acos construction modifies the background expansion through the bounded-clock non-vacuum kernel, rather than by adding an independent clustering component. In the standard subhorizon growth equation, the modification therefore enters through $H(z)$ and the usual matter Poisson source term, with no perturbation-level sound speed to specify (Sec. II D). The shape of the resulting $f\sigma_8(z)$ trajectory is fixed at any given s_0 ; the overall amplitude is set by the usual $\sigma_{8,0}$ normalization. A consistency check at the SN+BAO best fit is presented in Sec. VI C, giving $\Delta\chi^2_{\text{RSD}} < 0.3$ in the diagonal-error consistency check across the data-allowed parameter range.

Current combined data (Pantheon+ and DESI DR2 BAO) constrain $s_0 < 0.19$ at 95% CL, consistent with

the Λ CDM limit. Λ cos matches Λ CDM at $\Delta\chi^2 = +0.11$ across the SN+BAO dataset. Adding compressed Planck distance priors shifts the preferred Ω_Λ , a background-distance consistency issue that is removed when Ω_Λ is freed ($\Delta\chi^2 = +0.28$ at equal parameter count). The parity with Λ CDM at current precision is a feature of the model’s design: the fiducial Λ CDM limit is recovered exactly as $s_0 \rightarrow 0$, so agreement with existing data is expected. The $(1+z)^1$ term becomes detectable only at the $\sim 1\%$ precision level in $H^2(z)$, which is where Λ cos and Λ CDM diverge. The model is a minimal one-parameter deformation of the fiducial flat Λ CDM expansion history, with a specific falsifiable signature (the negative $(1+z)^1$ term) testable by next-generation BAO measurements.

VIII. CONCLUSIONS

A non-phantom expansion history can produce apparent phantom crossings under standard two-parameter templates. Λ cos, a one-parameter bounded-clock deformation of fiducial flat Λ CDM, satisfies $w_{\text{eff}}(z) > -1$ at all redshifts under the fiducial-matter diagnostic split, while matching the Pantheon+ and DESI DR2 BAO distance-redshift relation at $\Delta\chi^2 = +0.11$ relative to flat Λ CDM. Current data constrain the model’s single new parameter to $s_0 < 0.19$ at 95% confidence, and the predicted negative $(1+z)^1$ contribution to $H^2(z)$ is a target for next-generation BAO measurements.

Appendix A: Clock Exponent Selection

Three alternative clock rates were tested against the joint Pantheon+ + DESI DR2 BAO dataset using the same MCMC setup as the primary Λ cos fit (Sec. VB): identical priors on $H_0 r_d$ and M_B , identical sampler configuration, identical likelihood construction.

*Model B saturates at the s_0 prior floor. The likelihood is monotonic toward $s_0 \rightarrow 0$ under the $(1+z)^2$ scaling, so the reported value reflects the boundary, not a posterior peak.

[†]For Λ cos (model D), the chain argmax is $s_0 = 0.008$; the likelihood surface is flat near $s_0 = 0$, consistent with the Λ CDM limit. The posterior summary is reported in Sec. VB.

Acceptance fractions: A 0.65, B 0.62, C 0.65; all chains converged with $\tau_{\text{max}} < 50$ for all parameters.

Each integer alternative fails for a distinct reason. Model A’s $(1+z)^1$ scaling is too soft to reproduce matter dilution; the supernova sector tolerates this within $\Delta\chi^2_{\text{SN}} \approx 55$, but the BAO sector adds a 164 penalty as the high-redshift bins ($z = 1.32, 1.48, 2.33$) reject the soft scaling. Model B saturates against the s_0 floor because the $(1+z)^2$ scaling offers no improvement over Λ CDM at any positive s_0 ; the BAO sector dominates the rejection at $\Delta\chi^2_{\text{BAO}} \approx 1747$. Model C’s $(1+z)^0$ scaling at high redshift is the most dramatic failure: with no decay of the

matter-like term, both sectors reject the model ($\Delta\chi^2_{\text{SN}} \approx 980$, $\Delta\chi^2_{\text{BAO}} \approx 6300$).

Within the power-law clock family, the exponent $n = -1/2$ is selected analytically by three-dimensional matter dilution ($\rho_m \propto S^{-3}$) through the Friedmann square root, giving the non-vacuum scaling $\mathcal{K}^{1/2} \propto S^{-3/2}$ at high z and the exact two-term non-vacuum kernel of Sec. IIC at all z . The empirical fits confirm the analytic selection: among the alternatives tested, only $n = -1/2$ is viable.

DATA AND CODE AVAILABILITY

The MCMC chains, analysis scripts, and figure-generation code that reproduce all results in this paper are publicly available at github.com/dmobius3/lambdacos and archived at Zenodo (DOI:10.5281/zenodo.19798852). The Pantheon+ supernova compilation [2] is available at pantheonplussh0es.github.io and the DESI DR2 BAO measurements [1] at data.desi.lbl.gov.

TABLE X. Clock-exponent comparison from Appendix A: integer alternatives versus the matter-era selection $n = -1/2$.

Model	n	High-z scaling	Best-fit s_0	$H_0 r_d$ (km/s)	χ^2_{SN}	χ^2_{BAO}	χ^2_{total}	$\Delta\chi^2$ vs ΛCDM
A (proper time)	0	$(1+z)^1$	0.823	9279	1814.7	176.4	1991.1	+218.7
B (conformal)	-1	$(1+z)^2$	0.001*	8555	1834.5	1759.9	3594.4	+1821.9
C (symmetric)	+1	$(1+z)^0$	0.962	9220	2737.1	6312.0	9049.1	+7276.6
D (Acos)	-1/2	$(1+z)^{3/2}$	0.008[†]	10025	1759.0	13.5	1772.6	+0.11
ΛCDM (baseline)	—	—	($\Omega_m = 0.312$)	10044	1759.9	12.6	1772.5	0

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- [1] DESI Collaboration, arXiv:2503.14738 (2025).
[2] D. Brout *et al.*, *Astrophys. J.* **938**, 110 (2022).
[3] Planck Collaboration VI, *Astron. Astrophys.* **641**, A6 (2020).
[4] M. Chevallier and D. Polarski, *Int. J. Mod. Phys. D* **10**, 213 (2001).
[5] E. V. Linder, *Phys. Rev. Lett.* **90**, 091301 (2003).
[6] A. Vikman, *Phys. Rev. D* **71**, 023515 (2005).
[7] Y.-F. Cai, E. N. Saridakis, M. R. Setare, and J.-Q. Xia, *Phys. Rep.* **493**, 1 (2010).
[8] M. Cortès and A. R. Liddle, arXiv:2504.15336 (2025).
[9] DESI Collaboration, arXiv:2503.14743 (2025).
[10] E. J. Barboza and J. S. Alcaniz, *Phys. Lett. B* **666**, 250 (2008).
[11] H. K. Jassal, J. S. Bagla, and T. Padmanabhan, *Mon. Not. R. Astron. Soc.* **356**, L11 (2005).
[12] E. V. Linder and D. Huterer, *Phys. Rev. D* **72**, 043509 (2005).
[13] A. Shafieloo, V. Sahni, and A. A. Starobinsky, *Phys. Rev. D* **80**, 101301 (2009).
[14] R. E. Keeley, A. Shafieloo, and W. L. Matthewson, arXiv:2506.15091 (2025).
[15] S. Weinberg, *Gravitation and Cosmology* (Wiley, New York, 1972).
[16] DESI Collaboration, *J. Cosmol. Astropart. Phys.* **2025**, 008, arXiv:2411.12021.
[17] W. Elbers *et al.*, arXiv:2503.14744 (2025).
[18] D. Forero-Sánchez *et al.*, arXiv:2602.18761 (2026).
[19] Euclid Collaboration, *Astron. Astrophys.* **662**, A112 (2022).
[20] S. Zhou *et al.*, *Mon. Not. R. Astron. Soc.* **521**, 2 (2023).